

Fuel Delivery App

Course: Software Engineering

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Section: [E]

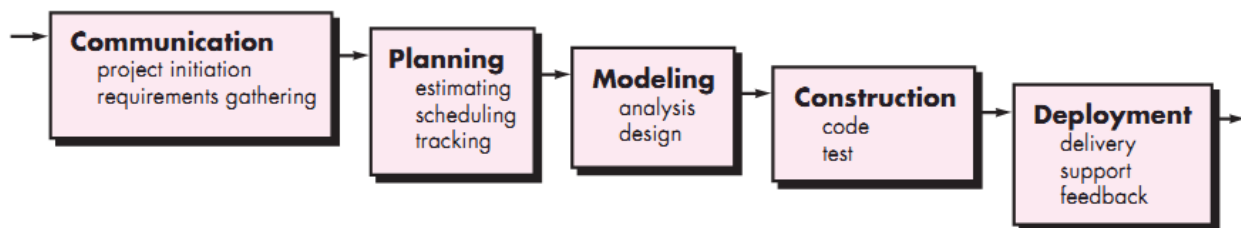
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FUEL DELIVERY APP

Model: Waterfall model

The waterfall model is a classical model used in system development life cycle to create a system with a linear and sequential approach. It is termed as waterfall because the model develops systematically from one phase to another in downward fashion. The waterfall approach does not define the process to go back to the previous phase to handle changes in requirement. The waterfall approach is the earliest approach that was used for software development.



The waterfall methodology uses a sequential or linear approach to software development. The project is broken down into a sequence of tasks, with the highest-level grouping referred to as *phases*. A true waterfall approach requires phases that are completed in sequence and have formal exit criteria, typically a sign-off by the project stakeholders.

A typical list of waterfall tasks would include:

- Scope and plan project
- Gather and document requirements
- Design application
- Develop application and perform unit tests
- Conduct system testing
- Perform UAT
- Fix application as appropriate
- Deploy application

The waterfall methodology is a formal process, with each phase comprising a list of detail tasks with accompanying documentation and exit criteria. Larger enterprises often require the use of SDLC methodology products, particularly in larger IT application projects. This is also the approach that SIs use when building IT applications for their customers, since budget, resources, deliverables, and scope must be managed on a very disciplined basis.

The advantages of the waterfall methodology are that:

- Requirements are completed early in the project, enabling the team to define the entire project scope, create a complete schedule, and design the overall application.
- It improves resource utilization because tasks can be split to be worked in parallel or grouped to leverage resource skills.
- It is a better application design because there is a more complete understanding of all the requirements and deliverables.
- The project status is more easily measured based on a complete schedule and resource plan.